Survival Modeling of Cirrhosis Patients Using Statistical Learning Methods

Dataset: Cirrhosis Patient Survival Prediction

We used a Kaggle dataset. It utilizes 17 clinical features for predicting survival state of patients with liver cirrhosis. The survival states include 0 = D (death), 1 = C (censored), 2 = CL (censored due to liver transplantation).
Cirrhosis results from prolonged liver damage, leading to extensive scarring, often due to conditions like hepatitis or chronic alcohol consumption. The data provided is sourced from a Mayo Clinic study on primary biliary cirrhosis (PBC) of the liver carried out from 1974 to 1984.

Goal:

The goal is to evaluate whether machine learning–based survival models can improve prediction accuracy compared to the traditional Cox proportional hazards model, and to determine if these models can meaningfully stratify patients into interpretable risk groups.

Research Questions

1. Which clinical variables significantly affect mortality risk?

2. Can ML survival models improve prediction compared to traditional Cox survival analysis?

3. How does the presence of a competing event (liver transplant) affect survival prediction?

4. Can we stratify patients into risk groups that meaningfully separate survival curves?

Study Design

- 70% training / 30% test split
- Fit multiple models on training
- Evaluate test C-index and visualize risk groups

Methods

- Cox Proportional Hazards Model: Baseline interpretable model
- LASSO / Elastic Net Cox (glmnet) :

  For feature selection and improved predictive stability.

- Random Survival Forests (RSF):

  A nonlinear, tree-based model capable of capturing complex interactions.

- Gradient-boosted survival models
- Risk Stratification:

  Use model-predicted risk scores to create high/medium/low risk groups and assess the separation of their Kaplan–Meier curves.

Traditional Approach: Cox Proportional Hazards Model

Key Findings:

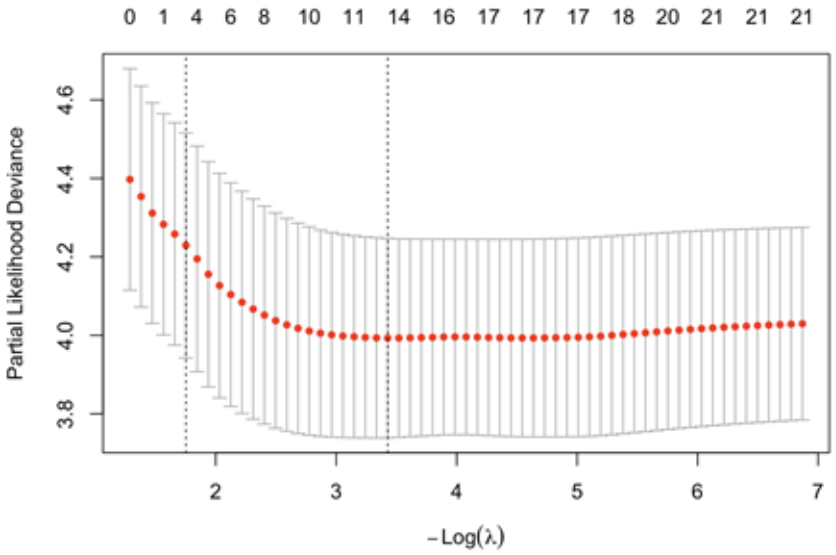
- Strongest predictor:
Ascites (HR = 3.06, p = 0.005)
- Protective factor: Albumin (HR = 0.53, p = 0.012)
- Laboratory markers: Bilirubin, Prothrombin highly significant
- Stage effect captured by continuous biomarkers

This forest plot shows the Cox model after stepwise selection. Patients with ascites have triple the mortality risk. Albumin is the only protective factor, with higher levels reducing death risk by 47%. Key laboratory markers like bilirubin and prothrombin are highly significant, while disease stage becomes non-significant once we control for these biomarkers, suggesting they capture the underlying disease severity.

Variable		N	Hazard ratio	p
Age		292	1.00 (1.00, 1.00)	<0.001
Ascites	N	280	Reference	
	Y	12	3.06 (1.41, 6.66)	0.005
Bilirubin		292	1.10 (1.06, 1.14)	<0.001
Albumin		292	0.53 (0.32, 0.87)	0.012
Copper		292	1.00 (1.00, 1.01)	0.009
SGOT		292	1.00 (1.00, 1.01)	0.013
Prothrombin		292	1.28 (1.10, 1.49)	0.001
Stage	1	16	Reference	
	2	69	1.85 (0.42, 8.15)	0.418
	3	111	2.78 (0.64, 12.11)	0.174
	4	96	4.31 (0.99, 18.72)	0.051

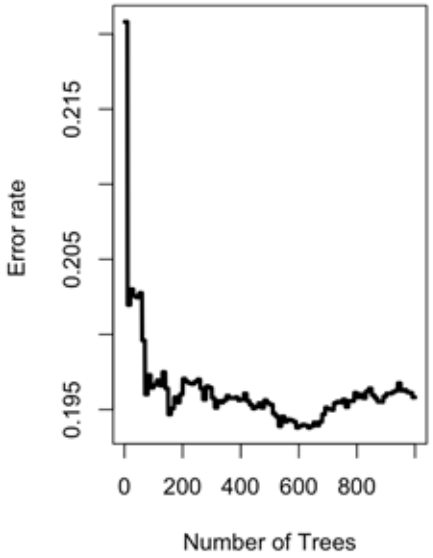
ML Models: Variable Importance & Model Performance

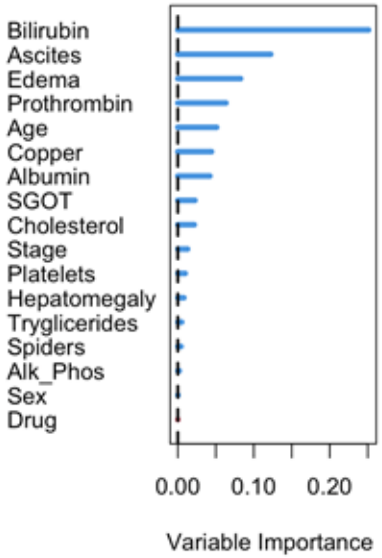
LASSO-Cox: Cross-Validation for Regularization



- Optimal λ: 13 non-zero coefficients
- C-index: 0.8574 (best among all models)
- Top predictors: Ascites, Albumin, Stage, Edema
- Excluded: Drug, Sex,Cholesterol, Tryglicerides
, Platelets (non-significant)
- Improvement over Cox: +0.3 % C-index

RSF: Variable Importance Plot

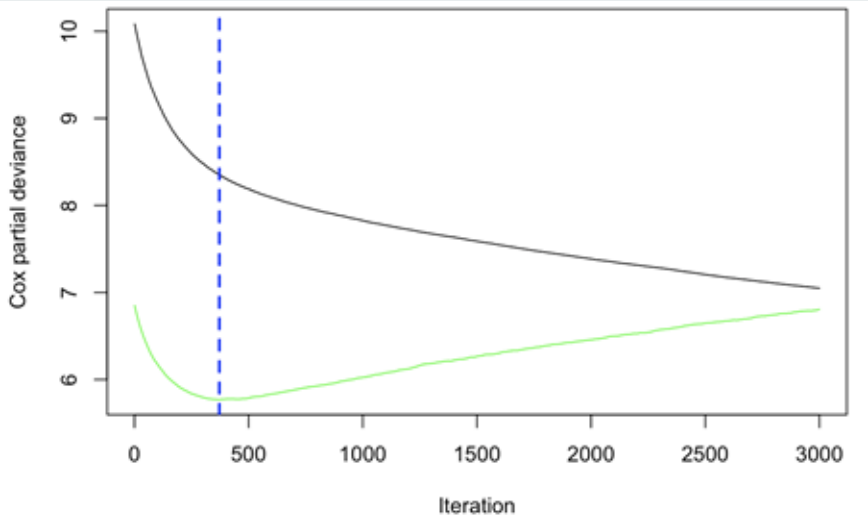




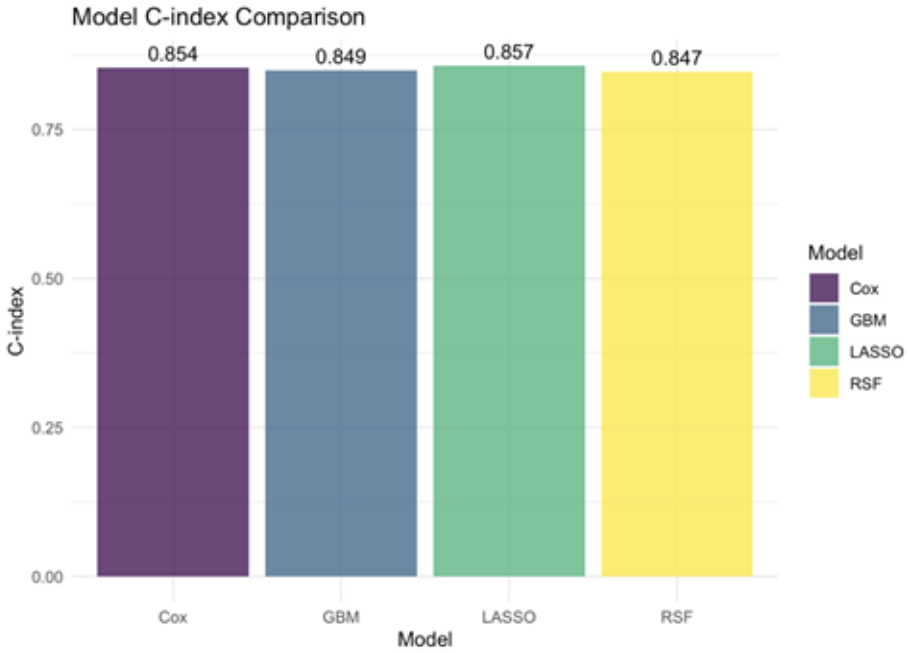
- Model converges at ~200 trees
- Bilirubin: Dominant predictor (2× others)
- Top 5: Bilirubin > Ascites > Edema > Prothrombin > Age
- Drug/Sex: Negligible importance
- Consistent with Cox/LASSO findings

Variable Importance & Model Performance

Gradient-boosted cross-validation plot

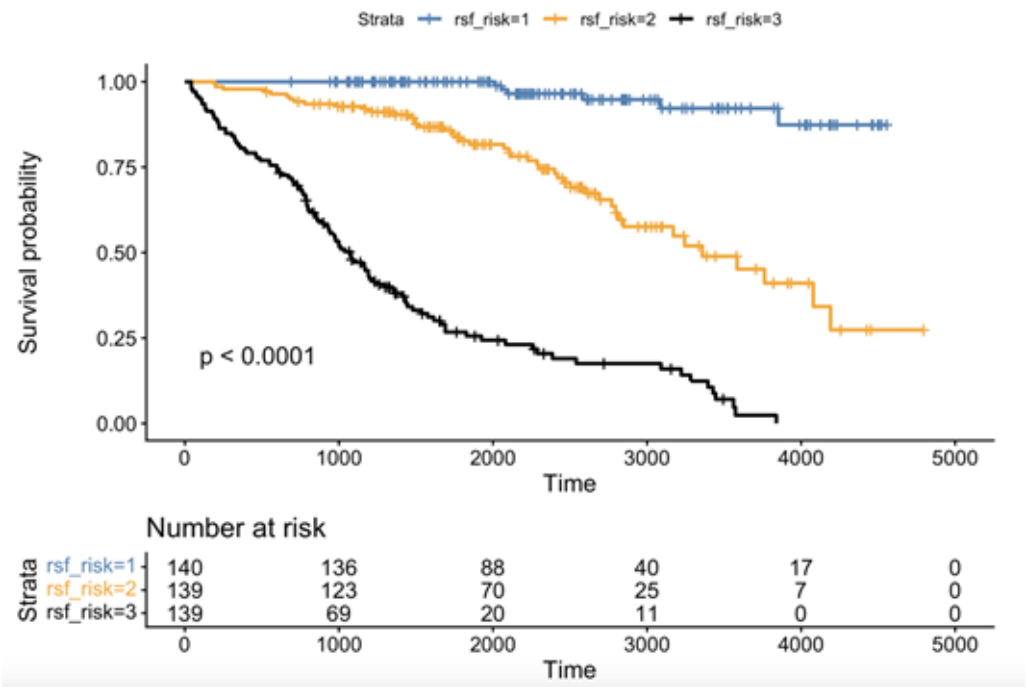


This plot shows our GBM model's cross-validation process. Green curve represents CV error across different numbers of trees. **Black Line** represents the model's **training error** (Cox partial deviance). We see the error drops rapidly initially, reaches a minimum around 500 trees (blue line), then slightly increases—indicating overfitting.



C-index: Measures the model's ability to correctly rank patients by risk.

LASSO Risk Stratification KM Curve



Risk Group Separation:

- High Risk (n=139): Median survival ~1000 days

- Medium Risk (n=139): Median survival ~2000 days

- Low Risk (n=140): 80%+ survival at 3000+ days

Clinical Implication:

Clear stratification enables personalized treatment planning and monitoring intensity

Conclusion

Machine learning models, particularly LASSO-Cox (C-index: 0.857), achieve marginal improvement over traditional Cox regression while providing automatic feature selection. All models identify liver function biomarkers (Bilirubin, Albumin, Prothrombin) as key predictors and successfully stratify patients into risk groups ($p < 0.001$).

These findings support the use of regularized statistical learning for survival prediction and emphasize that model interpretability remains crucial for clinical implementation.